

Control

Ziegler-Nichols Tuning.

> Time domain tuning

two methods :

Oscillation Approach

(a) Close-loop + p-controller

tune K_p , s.t. oscillate (i.e. marginally stable)

this gives a value of gain K_c

(b) Note K_c and period of oscillation T_c .

(c) The recommended gain settings for:

P: $K_p = 0.5 K_c$

PI: $K_p = 0.45 K_c$

$T_I = 0.833 T_c$

PID: $K_p = 0.6 K_c$

$T_I = 0.5 T_c$

$T_D = 0.125 T_c$

Open-loop Step Approach

(a). generate open-loop step response

(b) calculate slope R

lag time L

(c) The recommended gain settings :

$$P: K_p = 1/(CL)$$

$$PI: K_p = 0.9/(CL)$$

$$T_1 = 3.3L$$

$$PID: K_p = 1.2/(CL)$$

$$T_1 = 2L$$

$$T_0 = 0.5L$$